Homework 2

CS355/CE373 Database Systems

Fall 2024

Dhanani School of Science and Engineering

Habib University

Contents

1 Instructions

- The deadline for submitting this Homework is Monday, October 28th, 2024 by 11:59 PM.
- This homework must be submitted online via CANVAS.
- You are required to submit a **single sql** which contains solutions of all questions.
- Follow the same pattern as followed in the lab when answering the questions, i.e. comment on one line followed by its respective query.
- The sql file should be named HW_02_aa01234.sql where aa01234 will be replaced with your student id.
- *Files that don't follow the appropriate naming convention will not be graded.*

1.1 Marking scheme

This Homework will be marked out of 100.

- Correctness(85 points - 5 for each):
- Efficiency(5 points): Queries are written efficiently avoiding unnecessary sub-queries or joins, and minimizing resource usage.
- Clarity (5 points): Queries are understandable and logically structured that follow coherent flow.
- Comments (5 points): Brief explanations provided in the code to describe the purpose or logic behind key parts of the query.

1.2 Submission Guidelines

- You may use SQL Server Management Studio or any online SQL tool to run your queries
- **No handwritten submissions will be accepted**
- If you make any assumptions regarding the scenario, clearly state them in comments in your SQL file. **Your assumptions should not contradict the actual scenario.**

1.3 Late submission policy

Refer to the course syllabus for the late submission policy.

1.4 Use of AI

Taking help from any AI-based tools such as ChatGPT is strictly prohibited and will be considered plagiarism.

1.5 Viva

Course staff may call any group/student for Viva to explain their submission.

2 Objective

This assignment enables students to use SQL queries to run different functions on the given ERD.

3 ERD Description

An organization wants to store its employees' data using Information Systems. The system stores information about each worker with their respective departments. The system also stores information when a certain designation is assigned to a respective person. It is important to mention that system can store future designations as well: a designation can be affected in the future as well. The Bonus entity stores any bonus awarded to any employee.

The ERD Diagram for the Database is as shown in Figure 1.

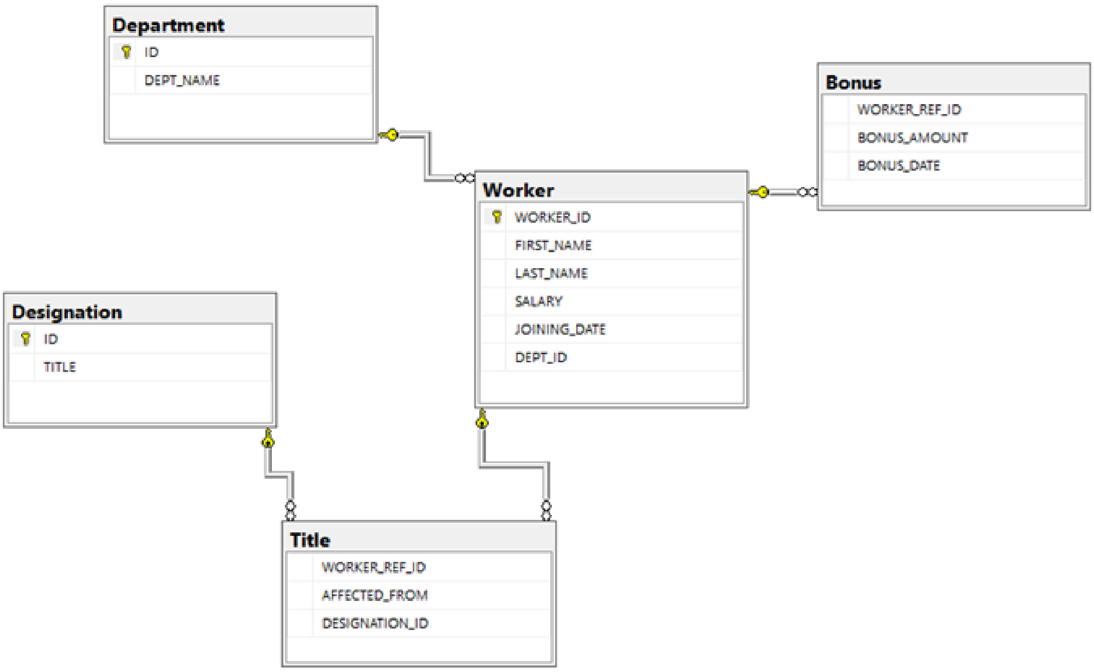

Figure 1: HW ERD

4 Task

First create a database called "Homework_02", and run the given DDL script file to create tables and enter values in them. Once that is successfully created, run and test the following queries:

5 Questions

1. **List the number of employees in each department.**
   Output: Dept_Name, No_Of_Workers.
   Result contains 3 rows.

2. **Display the employment history of worker 'Malik Zain'; i.e. list all his designations where current designation must be at the top and the first designation in the company must be at the bottom (ignore future designations).**
   Output: First_Name, Last_Name, Title, Affected_From.
   Result contains 3 rows.

3. **Display the total salary of all workers belonging to the department 'Admin'.**
   Output: Dept_Name, Salary.
   Result contains 1 row.

4. **List all workers with their salaries and department name. If a worker has not been assigned any department, then the result must show 'No Department' for those workers.**
   Output: First_Name, Last_Name, Salary, Dept_Name .
   Result contains 10 rows.

5. **List all workers with their joining dates (without time) and the total number of years of service in the company.**
   Output: First_Name, Last_Name, Joining_Date, Years.
   Result contains 10 rows.

6. **List all records of the 'Worker' entity.**
   Output: Worker_ID, First_Name, Last_Name, Salary, Joining_Date, Dept_ID
   Result contains 10 rows.

7. **List all the workers who do not have any designation and are not assigned to any department.**
   Output: Worker_ID, First_Name, Last_Name, Joining_Date
   Result contains 2 rows.

8. **List department names with the highest salary of the department.**
   Output: Dept_Name, Salary
   Result contains 3 rows.

9. **List all workers with the current designation order by First Name.**
   Output: First_Name, Last_Name, Title, Affected_From
   Result contains 8 rows.

10. **List all departments which have no workers assigned to them.**
    Output: Dept_Name
    Result contains 1 row.

11. **List workers joined on the same date - not necessary at the same time (An employee should not be listed if there is no other employee hired on the same date). You have to show new employees first.**
    Output: First_Name, Last_Name, Joining_Date (without time)
    Result contains 8 rows.

12. **List all workers who never got any bonuses. Your result must be sorted with respect to their names.**
    Output: First_Name, Last_Name, Joining_Date
    Result contains 4 rows.

13. **List names of all workers getting salaries more than the average salaries of their department.**
    Output: First_Name, Last_Name, Department, Salary
    Result contains 4 rows.

14. **List all workers with the total bonus they have received. If a worker has not received any bonus then it must show 0 for that worker. Your result must be sorted in a decreasing amount of total bonus.**
    Output: First_Name, Last_Name, Bonus_Total
    Result contains 10 rows.

15. **List all workers who received more than one bonus in a year (regardless of which year).**
    Output: First_Name, Department, Year (for which bonuses are awarded), No_Of_Bonuses (number of bonuses awarded in a specific year)
    Result contains 2 rows.

16. **List workers earning the second-highest salary in the organization.**
    Output: First_Name, Last_Name, Dept_Name, Salary
    Result contains 2 rows.

17. **List all workers who received bonuses in the year 2017 but not in the year 2018 along with the total bonus amount (regardless of year).**
    Output: First_Name, Last_Name, Dept_Name, Total_Bonus_Amount
    Result contains 3 rows.